PROVISIONAL PATENT APPLICATION

Counterfactual-Receipt Preservation System and Methods

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This specification is complete and self-contained. Companion applications are nonessential background only. Every example under PROPHETIC EXAMPLES is prospective; no universal cloak, complete multispectral concealment, address-denial mechanism, or qualified platform-scale system is represented as achieved.

Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.

CROSS-REFERENCE TO RELATED APPLICATIONS

This application belongs to a coordinated series concerning controlled boundary media, arrays, completion addresses, and transportation. A companion media application concerns general array hardware. The present invention is the counterfactual control architecture: acquisition of an absent-object reference, declaration of an observer or threat envelope, causal estimation of the live record that the protected object occludes, boundary synthesis, residual witnessing, joint adversarial classification, complete support accounting, and graceful degradation.

FIELD

Active and passive camouflage; adaptive optical, infrared, radio-frequency, acoustic, seismic, thermal, and multispectral boundary control; wavefront reconstruction; signature management; counterfactual scene estimation; observer-aware control; adversarial classification; and closed energy, momentum, heat, latency, and control accounting.

BACKGROUND

1. Transformation optics, scattering cancellation, phased arrays, adaptive camouflage, active noise control, wavefront reconstruction, conformal antennas, thermal management, and scene projection each suppress or replace selected signatures. Most are optimized per channel and under favorable geometry.
2. Suppressing object scattering alone produces an occlusion or dark region. A preservation system must also supply the observer-facing record that the absent object would have allowed from the background, including changes occurring after the object is present.

3. A prerecorded plausible scene fails under moving observers, parallax, unannounced background events, active interrogation, polarization, phase, out-of-band sensing, thermal load, or joint fusion of weak residuals.
4. The applicant treats cloaking as a bounded control problem: for a declared observer set, spectral and modality set, time interval, and test family, drive a controllable boundary so the measured object-present receipt belongs to the calibrated absent-object reference ensemble within a tolerance vector, while preserving the protected object and closing the support ledger.
5. This target is immediately useful short of universal invisibility. Boundary-cell metrology, adaptive aperture management, transient signature control, seam closure, protected sensing, background continuation, and adversarial fusion testing are practical products.

SUMMARY

A counterfactual-receipt preservation system (100; FIG. 1) includes a protected object or platform, distributed reference-acquisition sensors, a scene-state estimator, an observer/threat registry, controllable boundary cells, a control allocator, residual witness sensors, a joint classifier, a support-ledger recorder, a calibration memory, and a graceful-degradation controller.

The system freezes an absent-object reference ensemble and threat envelope; estimates the live counterfactual record using only causally available information; determines the residual between predicted reference and measured protected-scene receipt; allocates boundary controls from a measured response map or Jacobian; monitors joint residuals and support costs; and selects a lower declared objective or safe state when the full objective is infeasible.

Qualification advances from one boundary cell to static scene, dynamic scene, plural observers, multispectral fusion, platform integration, and adaptive adversary. The architecture explicitly distinguishes preservation, camouflage, and deception and stores which objective was commanded.

BRIEF DESCRIPTION OF THE DRAWINGS

- FIG. 1 — Counterfactual-receipt preservation system and closed control loop.
- FIG. 2 — Absent-reference, exposed-object, and controlled-boundary receipt ensembles.
- FIG. 3 — Observer and threat-envelope registry with locations, trajectories, apertures, waveforms, bands, and processing capabilities.
- FIG. 4 — Layered boundary cell with passive redirection, active reconstruction, sensing, state, and support interfaces.
- FIG. 5 — Causal scene estimator, lead time, prediction uncertainty, and unannounced-event test.
- FIG. 6 — Local receipt-control Jacobian, controllable subspace, and irreducible residual.
- FIG. 7 — Joint multispectral residual classifier and synchronized raw-data products.
- FIG. 8 — Complete support ledger for power, heat, momentum, angular momentum, latency, storage, recoil, emissions, and control traffic.

- FIG. 9 — Boundary-cell qualification fixture and adversarial trial matrix.
- FIG. 10 — Range progression from cell through platform and adaptive adversary.
- FIG. 11 — Graceful-degradation hierarchy and safe-state controller.
- FIG. 12 — Prophetic address-denial selector branch and independent discrimination gates.

DEFINITIONS

1. **Receipt** — the synchronized raw or calibrated observer-facing record of a declared modality, including amplitude, spectrum, phase, polarization, delay, direction, motion, parallax, temperature, pressure, vibration, or another stated observable.
2. **Absent-object reference ensemble** — repeated raw records and metadata acquired or independently reconstructed for the declared scene when the protected object and active boundary are absent, including natural variability and uncertainty.
3. **Exposed-object receipt** — the record with the object present and boundary preservation inactive.
4. **Controlled receipt** — the record with the boundary control active.
5. **Counterfactual receipt** — the live record predicted to have occurred under the absent-object condition, including lawful scene evolution hidden or disturbed by the protected object.
6. **Observer/threat envelope** — declared sensor positions and trajectories, apertures, waveforms, bands, polarizations, integration times, synchronization, processing capability, prior scene knowledge, and collaboration capability against which performance is claimed.
7. **Boundary cell** — an addressable boundary element capable of measured sensing, redirection, absorption, emission, phase control, polarization control, impedance control, vibration control, thermal transport, or another declared boundary operation.
8. **Boundary response map** — measured relation between cell commands, incident state, retained cell state, environment, and posted receipt, with uncertainty, latency, bandwidth, coupling, power, heat, and validity limits.
9. **Receipt residual** — a calibrated difference or statistical discrepancy between controlled and counterfactual receipts.
10. **Joint residual** — a synchronized multimodal record evaluated jointly rather than as a sum of independent thresholds.
11. **Preservation objective** — making the controlled record operationally indistinguishable from the absent-object reference within a frozen test family and tolerance.
12. **Camouflage objective** — reducing contrast or classification confidence without necessarily reproducing the absent-object record.
13. **Deception objective** — supplying a false but selected object, location, trajectory, or identity receipt.
14. **Controllable receipt subspace** — the span of local receipt changes producible by available boundary controls, estimated from a measured Jacobian or nonlinear map.

15. **Irreducible residual** — the component of exposed-object residual outside the controllable receipt subspace at the stated operating point.
16. **Support ledger** — synchronized power, energy, heat, linear and angular momentum, recoil, stored excitation, latency, computation, communications, emissions, and environmental-disturbance record.
17. **Graceful degradation** — selection of a stored lower objective when the requested preservation objective cannot be maintained, rather than unstable or conspicuous uncontrolled output.
18. **Address denial** — a prophetic boundary operation that suppresses an object-identifying completion independently of ordinary absorption, redirection, emission, scattering cancellation, or scene reconstruction.
19. **Qualified operating envelope** — the observer, modality, environment, scene, command, latency, damage, power, and duration region that passed the declared gates.
20. **Safe state** — a bounded passive, transparent, matched, quiescent, or declared signature state entered on invalidity or fault.

MATHEMATICAL AND CONTROL FRAMEWORK

Let **R0** denote the absent-object reference ensemble, **RP** the exposed-object record, and **RC(u)** the controlled record under boundary history **u(t)**. The preservation objective is

$$D_T(RC(u), R0) \leq \varepsilon$$

over declared projection domain Ω and frozen adversarial test family **T**. This is a bounded claim, not universal equality.

Because the platform occludes or disturbs part of the scene, the controller estimates

$$\hat{R}0(t+\delta \mid Z{\leq}t, M)$$

from physically available measurements **Z** and declared model **M**. Prediction uncertainty belongs in the residual.

Near an operating state, a measured Jacobian $Gu = \partial r/\partial u$ identifies the controllable receipt subspace. An approximate irreducible residual is

$$r\perp = (I - Gu\, Gu\dagger)\, rP.$$

The controller minimizes joint residual, power, heat, latency, and risk subject to causality, stability, custody, and ledger closure. The equations organize measurement; they do not derive an address-denial material law.

SYSTEM ARCHITECTURE (FIG. 1)

Reference sensors (110) observe background relations around or ahead of the protected object (120). The threat registry (130) stores the admitted observer envelope. Scene estimator (140) predicts the live absent-object record with covariance. Boundary array (150) comprises cells with passive and active

degrees of freedom. Allocator (160) maps desired receipt changes to bounded commands. Residual witnesses (170) observe edges, seams, support emissions, and protected channels. Joint classifier (180) evaluates raw synchronized records. Support ledger (185) closes physical accounts. Calibration memory (190) stores maps and provenance. Graceful-degradation controller (195) selects a lower objective or safe state.

OPERATING METHOD

The controller authenticates the map and threat registry; acquires or selects an absent-object ensemble; confirms exposed-object detectability; estimates the current counterfactual receipt; determines required correction and controllable projection; allocates boundary commands; applies them within latency, stability, power, and heat limits; records controlled receipts and support accounts; evaluates joint distinguishability; updates only within the validated adaptive region; and degrades or enters safe state on failure.

LAYERED SCOPE

- **N0:** cell response and raw synchronized receipt measurement.
- **N1:** single-channel static camouflage or preservation.
- **N2:** dynamic causal background continuation.
- **N3:** plural-observer and multispectral joint preservation.
- **N4:** integrated platform operation under adaptive adversary.
- **N5:** independently qualified address denial.

QUALIFICATION GATES

- **N1 reference custody:** absent-object records, variability, synchronization, and provenance are preserved;
- **N2 exposed detectability:** the frozen test detects the unprotected object with declared power;
- **N3 controlled residual:** the controlled record satisfies the frozen bound on held-out trials;
- **N4 causal novelty:** unannounced scene events and active interrogations exclude prerecorded replay;
- **N5 observer generalization:** held-out positions, trajectories, polarizations, bands, and waveforms satisfy the claimed envelope;
- **N6 joint fusion:** an independent red team cannot exceed the frozen classification bound using synchronized raw modalities;
- **N7 support closure:** power, heat, recoil, latency, emissions, and environmental disturbances are accounted;
- **N8 robustness:** seams, curvature, vibration, weather, fouling, damage, and calibration drift remain within limits or trigger graceful degradation;

- **N9 safety:** unstable emission, overheating, actuator runaway, observer-tracking loss, and corrupted reference cause a bounded state; and
- **N10 independent confirmation:** a second range, apparatus, scene, or team reproduces the claimed performance surface.

Address denial additionally requires that suppression persist after ordinary redirection, absorption, emission, cancellation, and reconstruction contributions are independently bounded and that near-match, phase-reversal, dummy-load, and observer-position controls survive.

EMBODIMENTS

E1—single optical boundary cell. A cell includes front and rear radiance sensors, controllable transmission/reflection/emission or a display/emitter, phase/polarization sensing where applicable, thermal sensing, and a controller. A calibrated background, adversarial target, moving camera, unannounced background event, and raw-data classifier form the qualification fixture.

E2—multispectral tile. Co-located visible, infrared, radio-frequency, and acoustic elements share timing but retain separate raw channels and support accounting. The tile controller prevents improvement in one modality from being hidden by migration into another.

E3—adaptive aperture and seam control. Cells surround windows, sensors, antennas, doors, and panel seams. The objective may be transient reduction during opening, startup, maneuver, or communication rather than continuous whole-platform preservation.

E4—platform array. A conformal array uses distributed reference acquisition, predictive state estimation, witness sensors, and hierarchical allocation. Damage isolates failed cells and recomputes a lower achievable envelope.

E5—prophetic address-denial cell (FIG. 12). A candidate selector varies a boundary state proposed to change eligibility of object-identifying completions while ordinary scattering and emission controls remain matched. The result is separately classified and is not inferred from successful reconstruction.

PROPHETIC EXAMPLES

P1—one-cell dynamic test. A 10–30 cm cell is placed before a calibrated changing background. Reference, exposed, dummy-powered, replay, and controlled trials are randomized. Cameras move over a declared angular range; polarization and thermal witnesses operate continuously. Success is a bounded held-out classification result, not visual impression.

P2—four-modality seam. A panel joint is instrumented in visible, long-wave infrared, radio frequency, and acoustic bands. The controller minimizes the joint residual subject to support limits and reports signature migration explicitly.

P3—platform progression. Qualification proceeds from cell to static panel, dynamic panel, multispectral assembly, instrumented unmanned platform, environmental range, and adaptive red team. No later scope is claimed from an earlier demonstration.

POAMS-NATIVE DIRECT-OPERATION EMBODIMENT

In the prophetic native embodiment, a visible object is an address stabilized by completed observer-facing boundary relations. The controller withholds the identifying participation while preserving the surrounding counterfactual receipts. The stronger address-denial mechanism remains an open physical selector and is not represented as achieved.

CLAIMS

1. A counterfactual-receipt preservation system comprising: reference-acquisition sensors configured to acquire scene records associated with absence of a protected object; an observer or threat registry storing an admitted sensor and processing envelope; a causal scene estimator configured to estimate a live counterfactual receipt and uncertainty from physically available measurements; a plurality of controllable boundary cells on or around the protected object; a calibration memory storing boundary response maps with latency, bandwidth, coupling, power, heat, uncertainty, and validity limits; a control allocator configured to command the boundary cells toward the counterfactual receipt; residual witness sensors; a joint classifier configured to evaluate synchronized raw controlled receipts against an absent-object reference ensemble; a support-ledger recorder; and a graceful-degradation controller configured to select a stored lower objective or safe state when a requested preservation objective is outside a qualified operating envelope.
2. The system of claim 1, wherein the boundary cells provide at least two operations selected from sensing, redirection, transmission, absorption, emission, phase control, polarization control, impedance control, vibration control, and thermal transport.
3. The system of claim 1, wherein the admitted envelope identifies observer positions or trajectories, apertures, waveforms, spectral bands, polarizations, integration times, synchronization, and processing capabilities.
4. The system of claim 1, wherein the causal scene estimator supplies a covariance or uncertainty bound and the allocator includes prediction error in the receipt residual.
5. The system of claim 1, wherein the calibration memory stores a measured local receipt-control Jacobian or nonlinear response map and the allocator identifies a controllable receipt subspace and an irreducible residual.
6. The system of claim 1, wherein the residual witnesses include edge, seam, thermal, recoil, out-of-band emission, actuator-noise, and latency witnesses.
7. The system of claim 1, wherein the joint classifier uses at least two synchronized modalities selected from visible, infrared, radio-frequency, radar, lidar, acoustic, seismic, magnetic, chemical, and navigation records.

8. The system of claim 1, wherein the support ledger records power, stored and dissipated energy, heat transport, linear and angular momentum, recoil, computation, communications, latency, and environmental disturbance.
9. The system of claim 1, wherein stored objectives separately identify counterfactual preservation, camouflage, and deception and the controller reports which objective is active.
10. The system of claim 1, wherein graceful degradation prioritizes declared receipts and prevents unstable or conspicuous actuator output when all residual constraints cannot be satisfied.
11. A method of preserving a counterfactual observer-facing receipt, comprising freezing an observer or threat envelope and a test family; acquiring an absent-object reference ensemble; confirming exposed-object detectability; estimating a live counterfactual receipt using only causally available measurements; determining a correction within a measured controllable receipt subspace; commanding boundary cells subject to latency, stability, power, heat, and custody limits; recording synchronized raw controlled receipts and a support ledger; testing joint distinguishability against the absent-object ensemble on held-out trials; and selecting a lower objective or safe state when a validity condition fails.
12. The method of claim 11, further comprising introducing an unannounced background event or active interrogation not available in a prerecorded replay and including the resulting prediction error in the evaluation.
13. The method of claim 11, further comprising testing held-out observer positions, trajectories, polarizations, bands, or waveforms inside the claimed envelope.
14. The method of claim 11, further comprising supplying synchronized raw data to an independent red-team classifier and qualifying the operation only when its detection or identification performance remains within a frozen bound.
15. The method of claim 11, further comprising measuring performance after a seam fault, cell failure, fouling, weather exposure, vibration, or damage and shrinking the qualified envelope accordingly.
16. A boundary-cell qualification apparatus comprising a calibrated changing background, an adversarial object, one or more boundary cells, movable observer sensors, polarization and phase instruments, thermal and recoil witnesses, a randomized-state controller, and a raw-data classifier configured to compare absent-object, exposed-object, dummy-powered, replay, and active-control trials.
17. The apparatus of claim 16, wherein the changing background includes an event whose timing or content is withheld from the boundary controller until physically observable.
18. The apparatus of claim 16, wherein the controller writes a cell response map or null map with measured angular, spectral, phase, polarization, latency, power, heat, and damage dependence.
19. A multispectral signature-control tile comprising co-located boundary elements for at least two modalities, a common timing reference, separate raw receipt channels, a local support ledger, and a controller configured to inhibit an operating point that reduces one residual while causing another modality to exceed a stored migration limit.

20. A platform-scale system comprising a plurality of tiles of claim 19, distributed reference acquisition, hierarchical control allocation, seam and edge witnesses, and a damage controller configured to isolate a failed tile and recompute a reduced qualified envelope.
21. A prophetic address-denial apparatus comprising a protected boundary, a controllable selector state, matched ordinary redirection, absorption, emission, cancellation, and reconstruction controls, observer-facing receipt sensors, and a controller configured to test whether the selector state suppresses an object-identifying completion independently of the matched ordinary controls.
22. A method of qualifying the apparatus of claim 21, comprising randomizing active selector, near-match, phase-reversal, dummy-load, ordinary-control, and observer-position states; measuring support and observer-facing receipts; bounding ordinary scattering, absorption, emission, redirection, cancellation, and reconstruction contributions; and reporting an independently confirmed address-denial response, an ordinary mechanism, an artifact, or a null.

ABSTRACT

A counterfactual-receipt preservation system acquires an absent-object reference ensemble, declares an observer or threat envelope, causally estimates the live scene record hidden or disturbed by a protected object, and commands mapped boundary cells toward that record. Residual witnesses, synchronized multimodal raw data, a joint adversarial classifier, and a complete support ledger measure performance, signature migration, latency, power, heat, recoil, emissions, and environmental disturbance. A graceful-degradation controller selects a stored lower objective or safe state when full preservation is infeasible. Qualification progresses from one cell through dynamic scenes, plural observers, multispectral fusion, platform integration, damage, and adaptive red-team tests. Optical, infrared, radio-frequency, acoustic, seam-control, transient-signature, and platform embodiments are disclosed. A separately gated prophetic address-denial branch tests suppression beyond ordinary redirection, absorption, emission, cancellation, or reconstruction; no universal cloak is represented as achieved.

Counterfactual-receipt preservation system and closed control loop

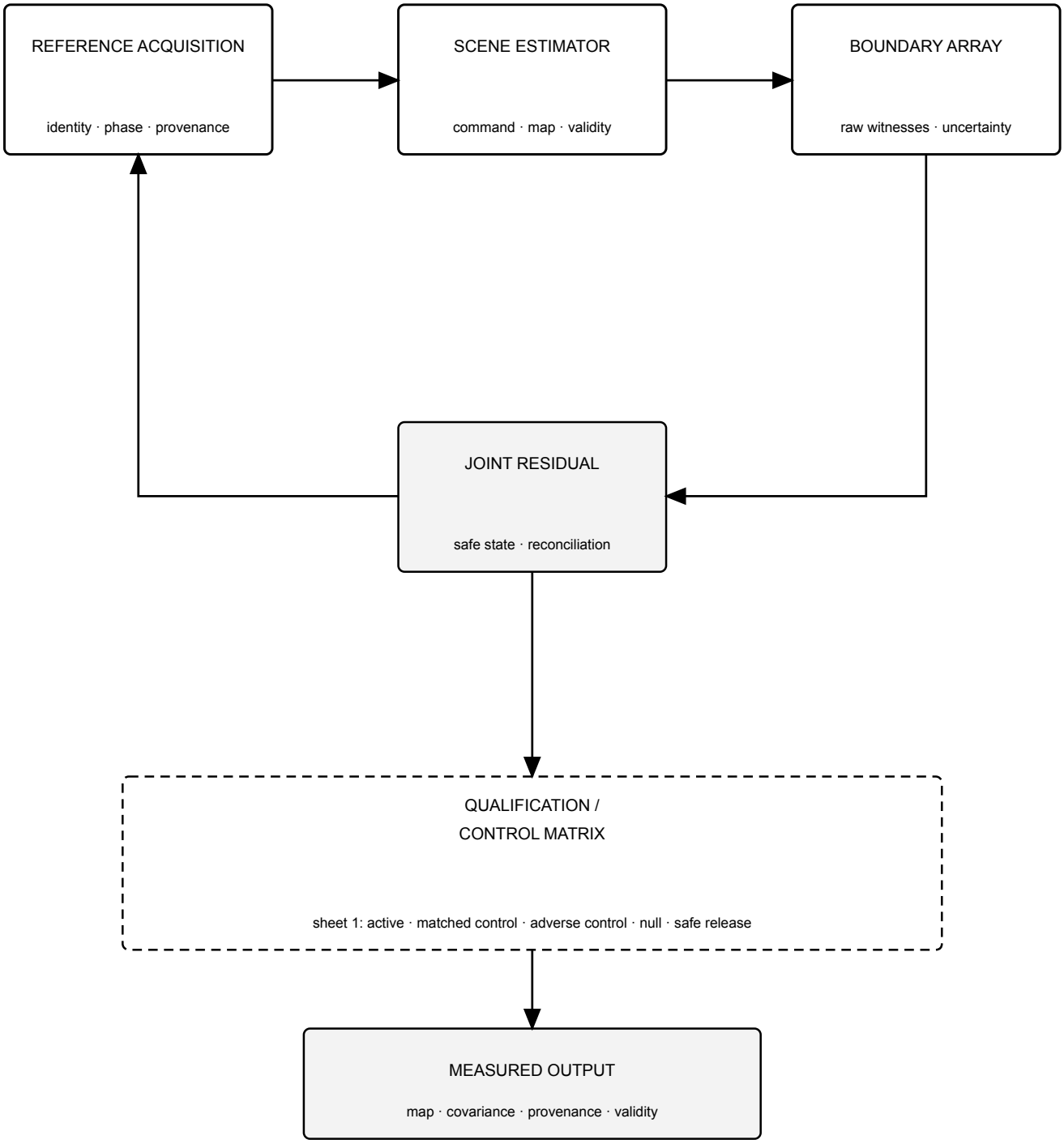


FIG. 1
Counterfactual-receipt preservation system and closed control loop.

Absent-reference, exposed-object, and controlled-boundary receipt ensembles

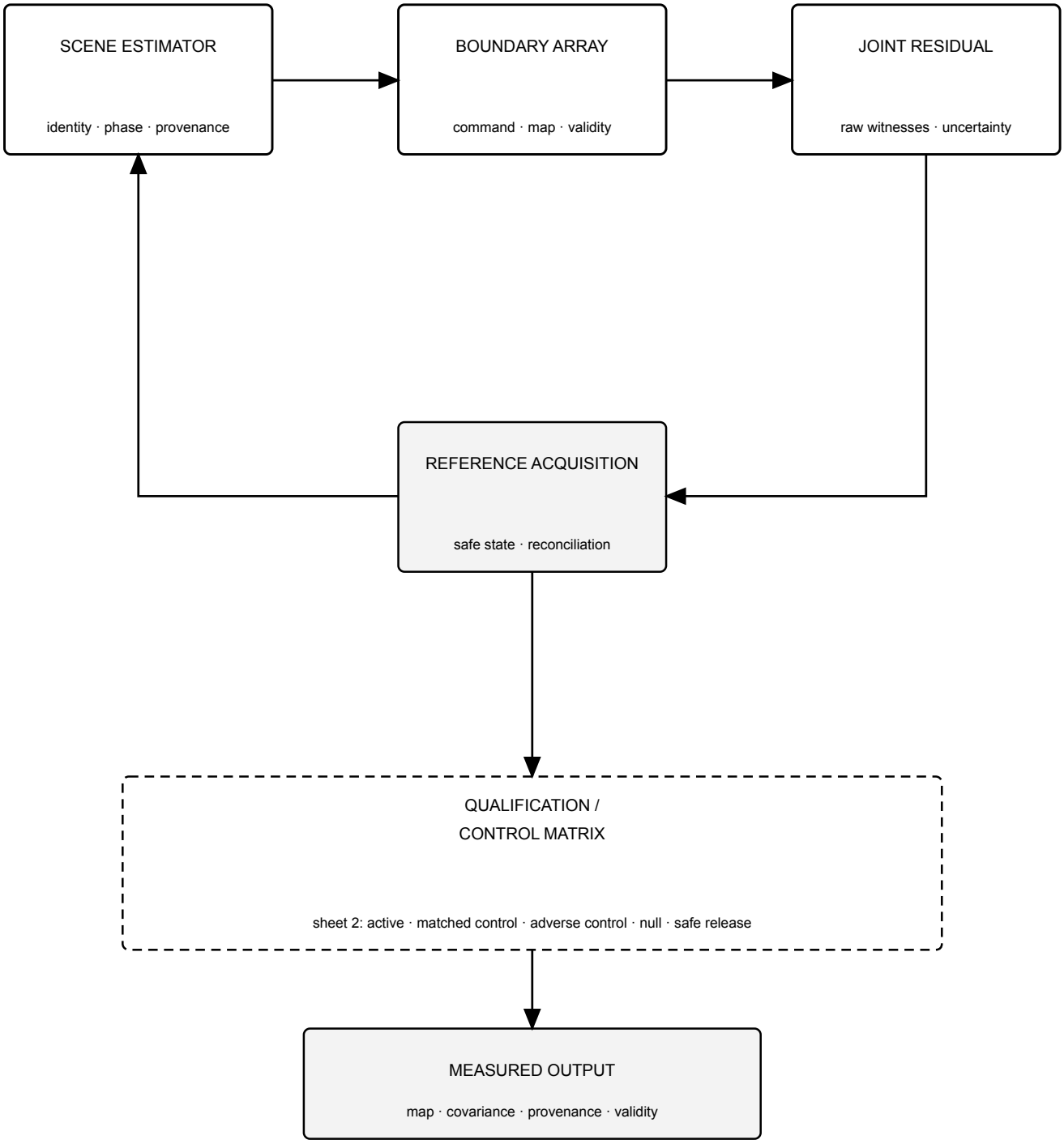


FIG. 2
Absent-reference, exposed-object, and controlled-boundary receipt ensembles.

Observer and threat-envelope registry with locations, trajectories, apertures, waveforms, bands, and processing capabilities

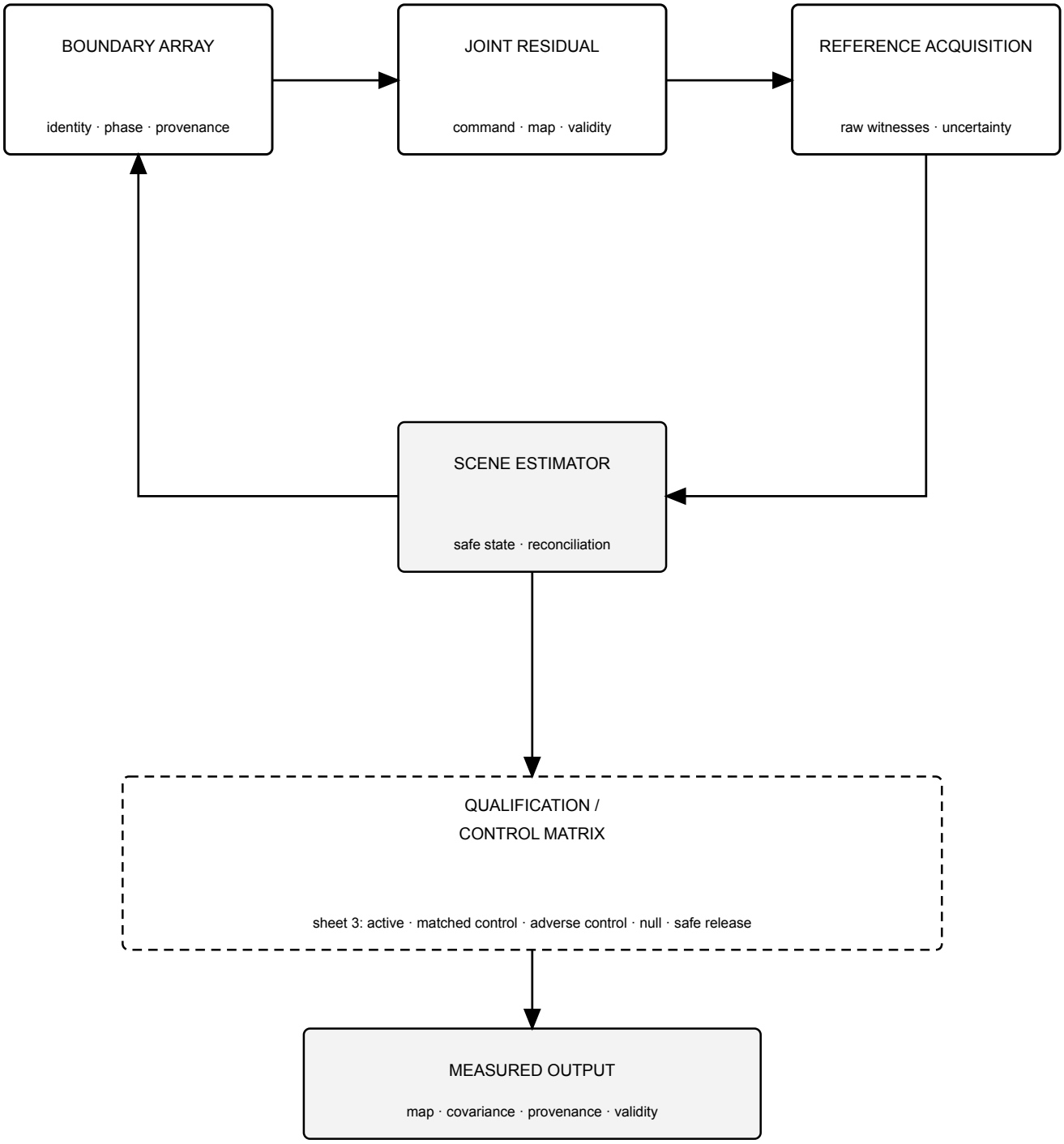


FIG. 3

Observer and threat-envelope registry with locations, trajectories, apertures, waveforms, bands, and processing capabilities.

Layered boundary cell with passive redirection, active reconstruction, sensing, state, and support interfaces

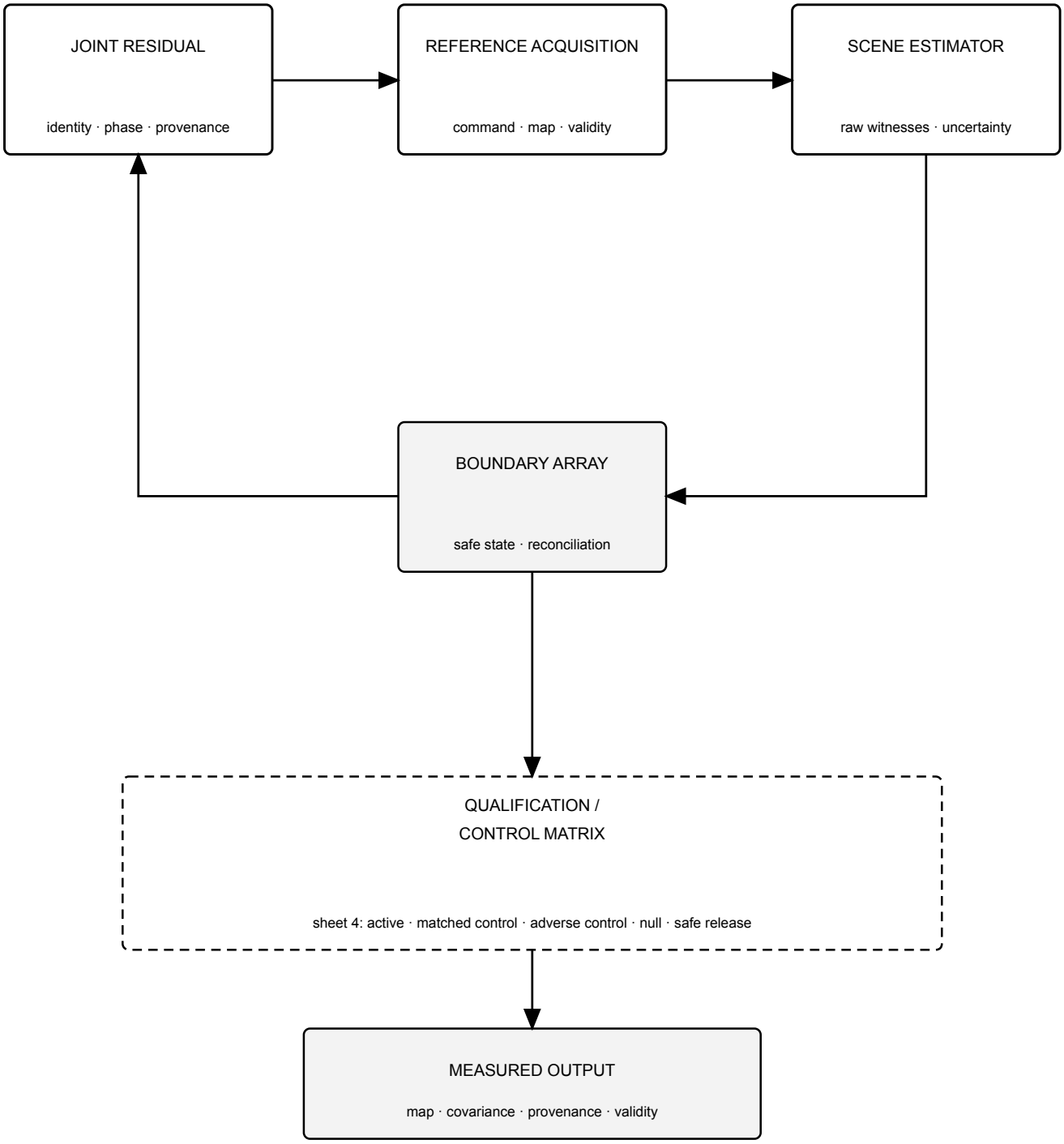


FIG. 4
Layered boundary cell with passive redirection, active reconstruction, sensing, state, and support interfaces.

Causal scene estimator, lead time, prediction uncertainty, and unannounced-event test

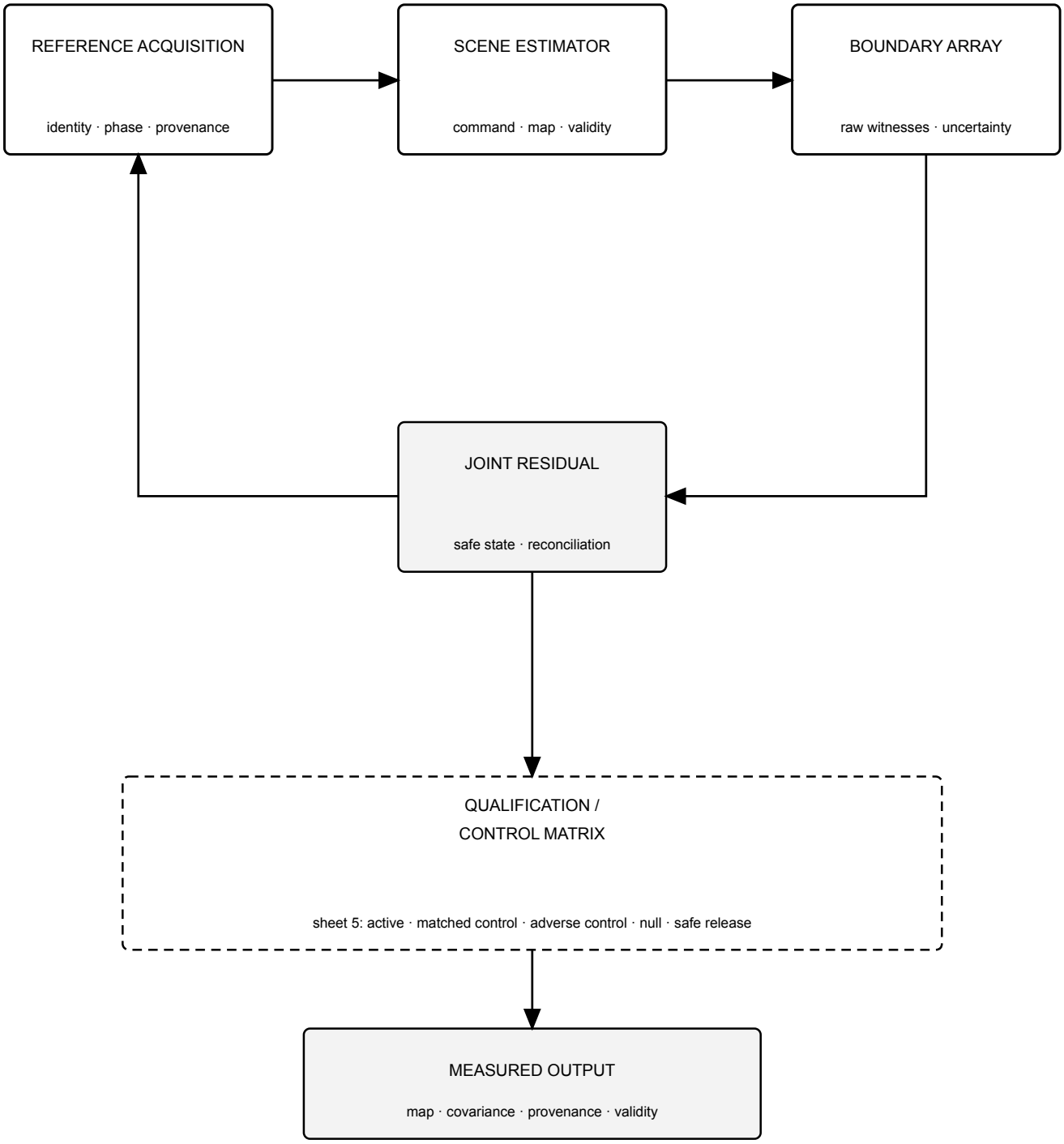


FIG. 5
Causal scene estimator, lead time, prediction uncertainty, and unannounced-event test.

Local receipt-control Jacobian, controllable subspace, and irreducible residual

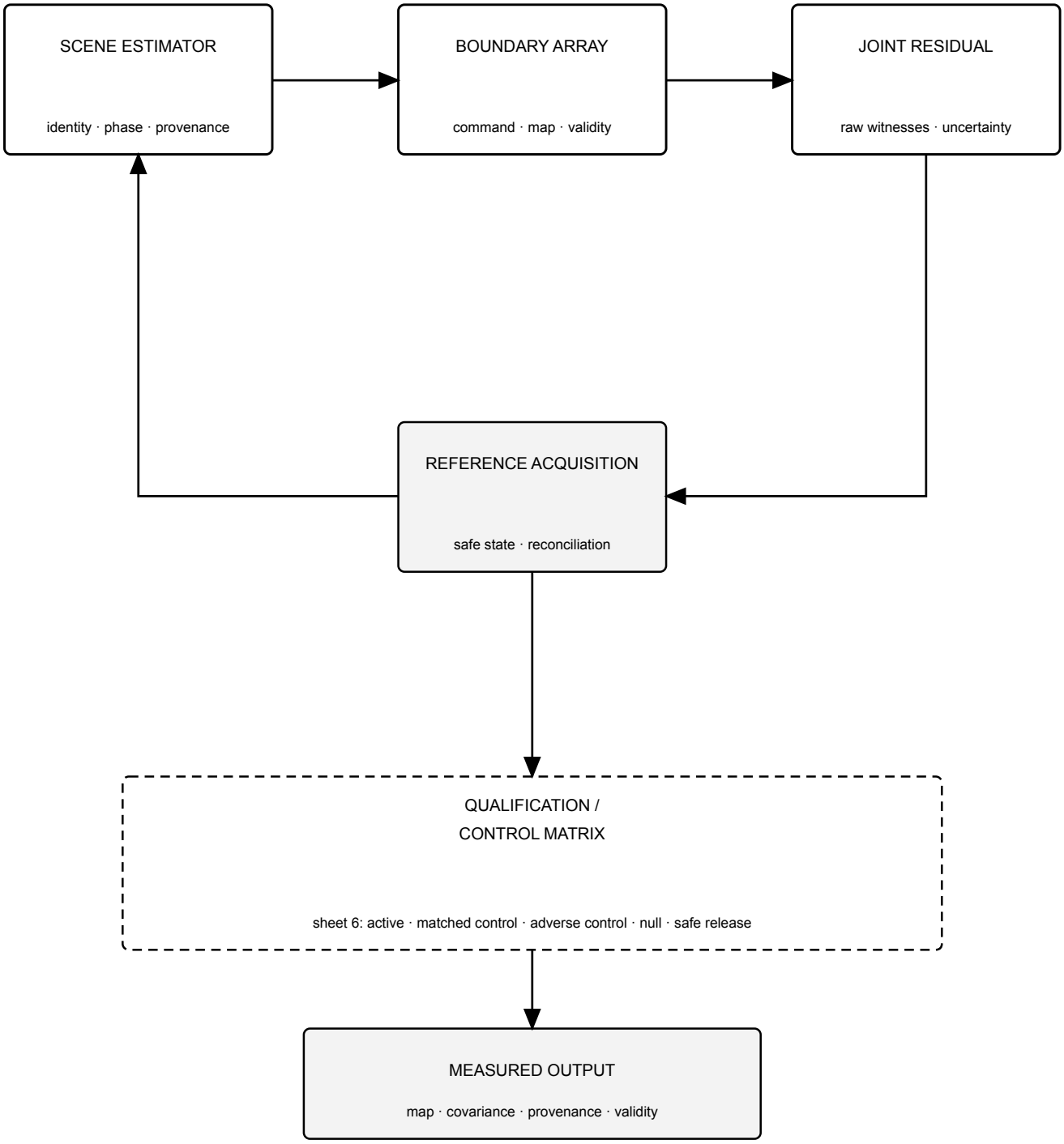


FIG. 6
Local receipt-control Jacobian, controllable subspace, and irreducible residual.

Joint multispectral residual classifier and synchronized raw-data products

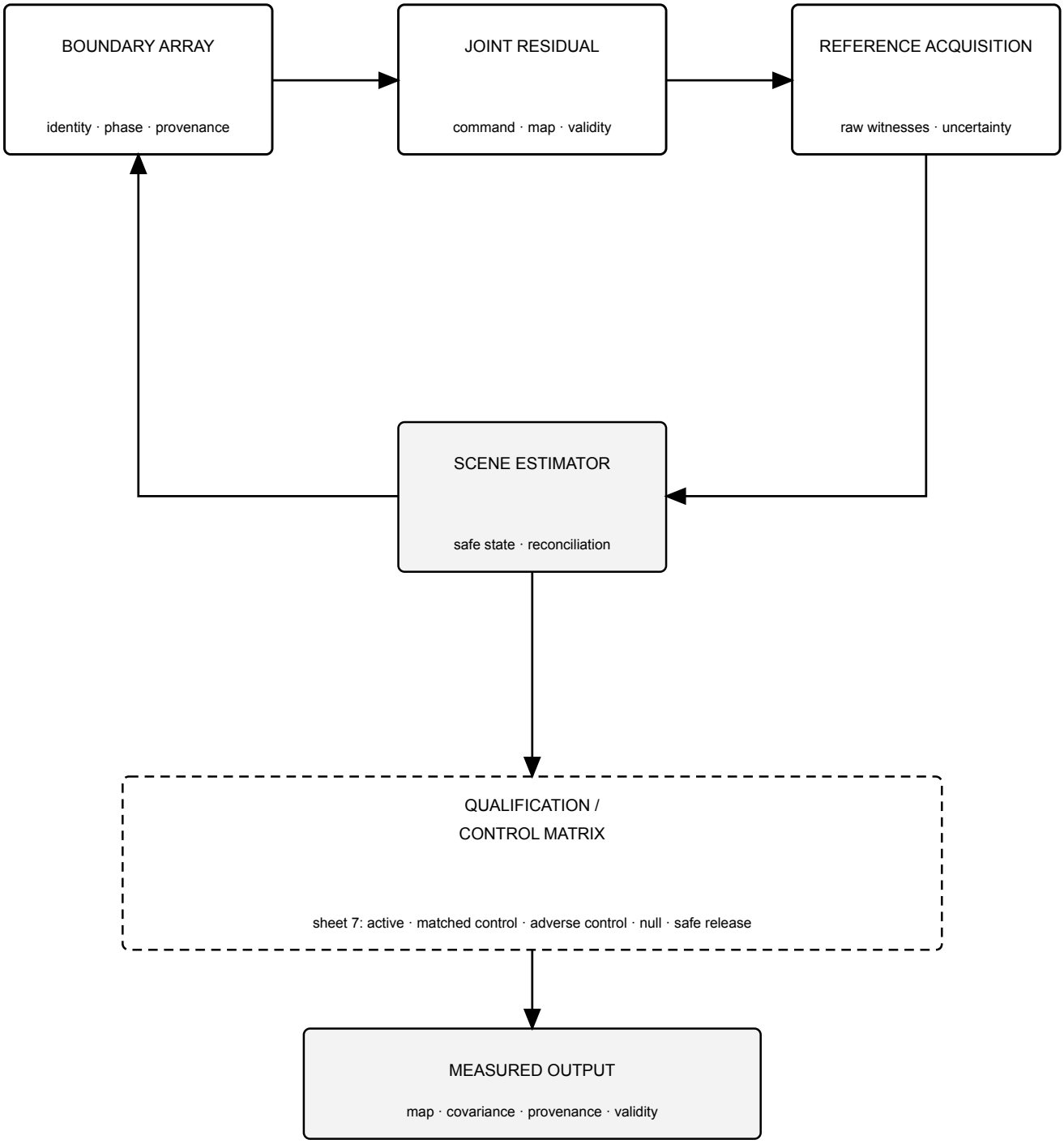


FIG. 7
Joint multispectral residual classifier and synchronized raw-data products.

Complete support ledger for power, heat, momentum, angular momentum, latency, storage, recoil, emissions, and control traffic

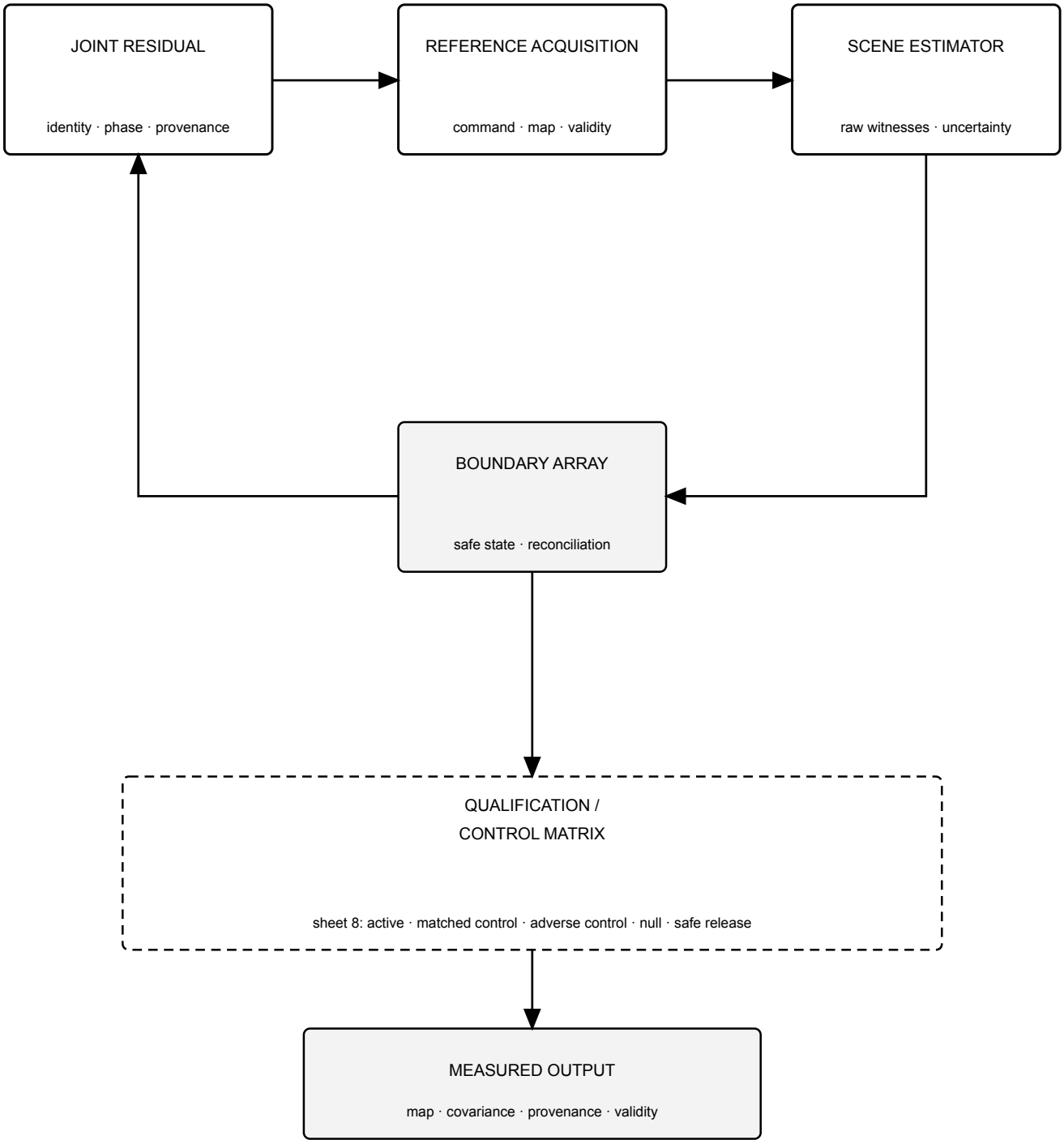


FIG. 8

Complete support ledger for power, heat, momentum, angular momentum, latency, storage, recoil, emissions, and control traffic.

Boundary-cell qualification fixture and adversarial trial matrix

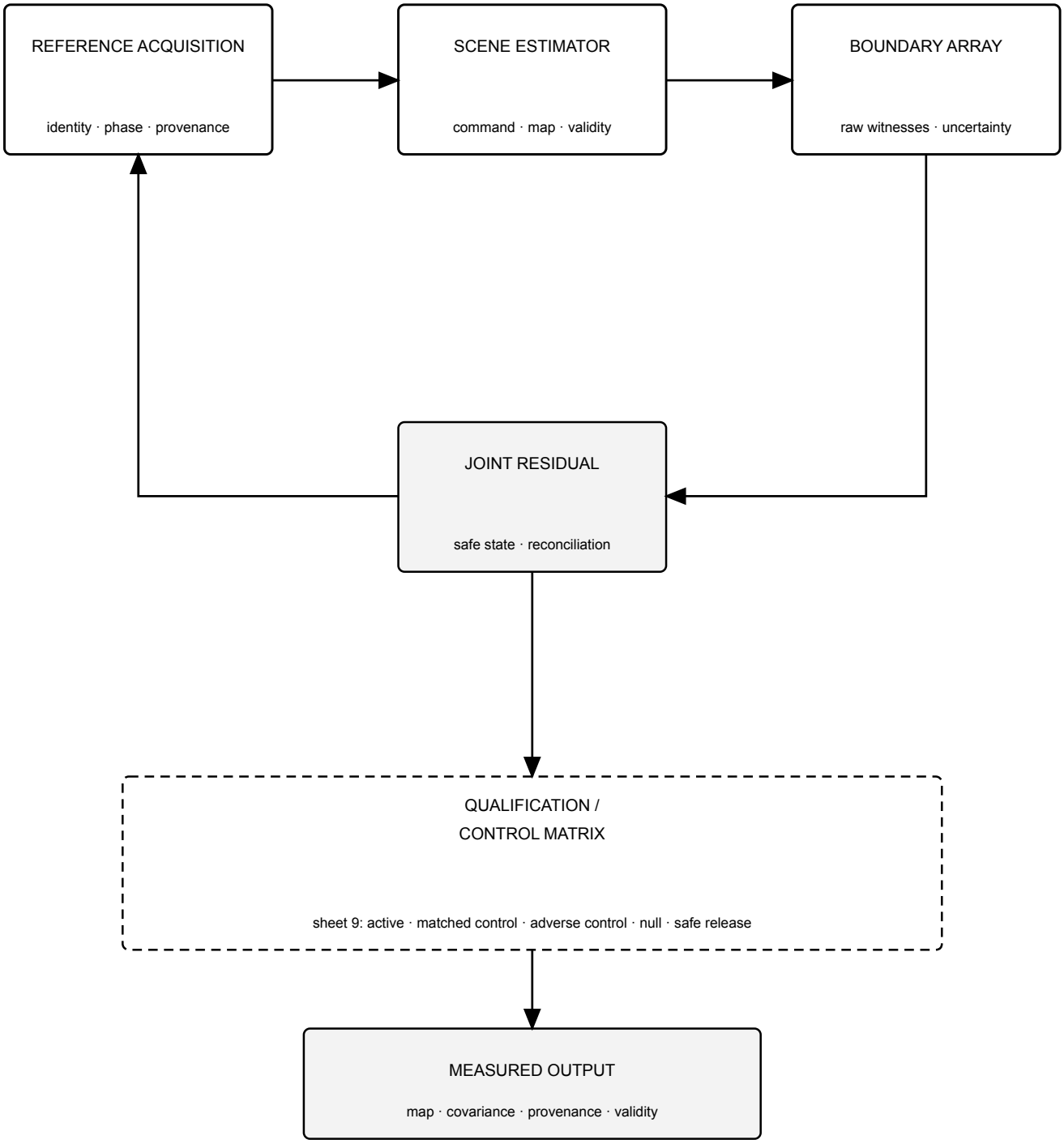


FIG. 9
Boundary-cell qualification fixture and adversarial trial matrix.

Range progression from cell through platform and adaptive adversary

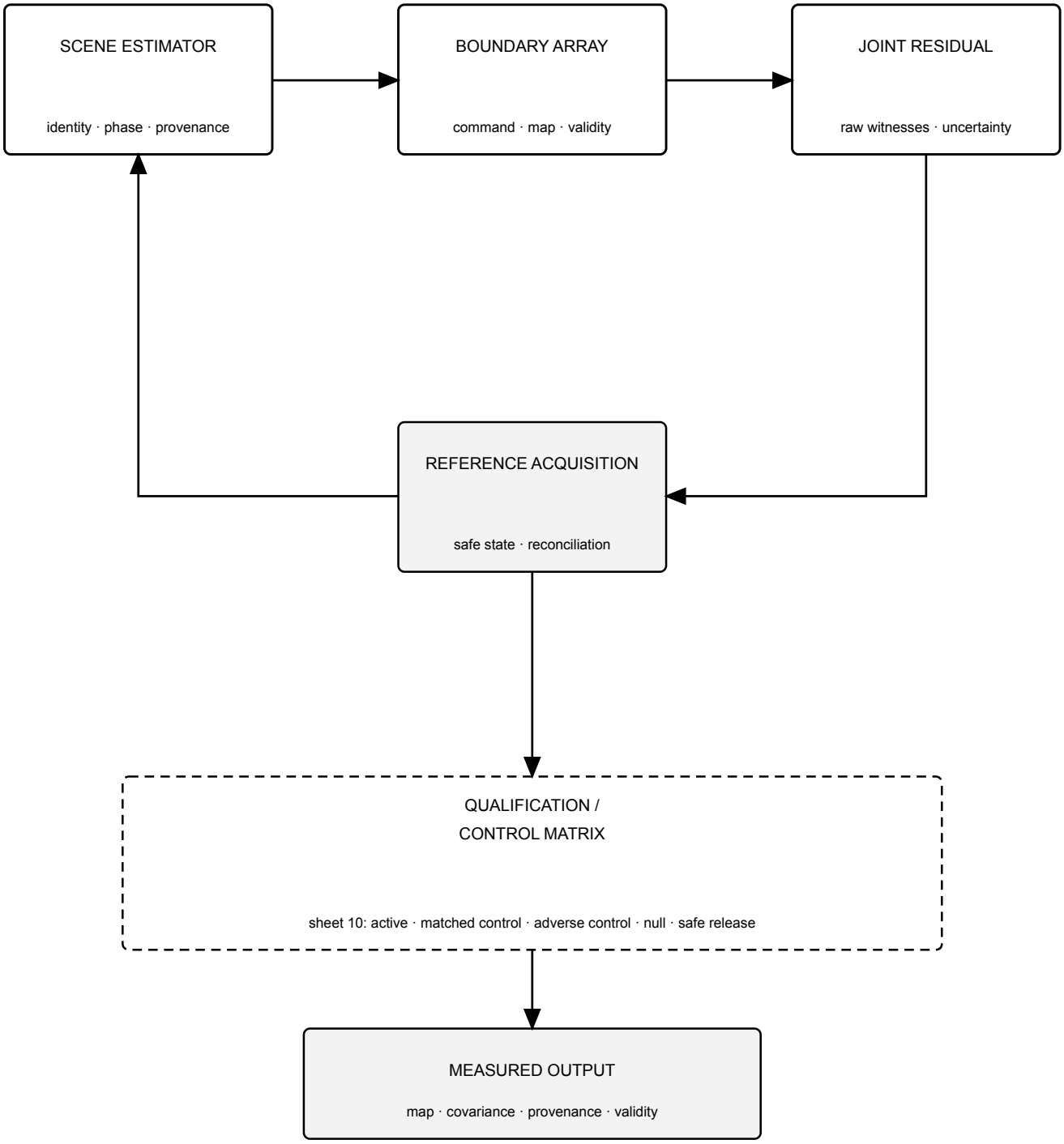


FIG. 10
Range progression from cell through platform and adaptive adversary.

Graceful-degradation hierarchy and safe-state controller

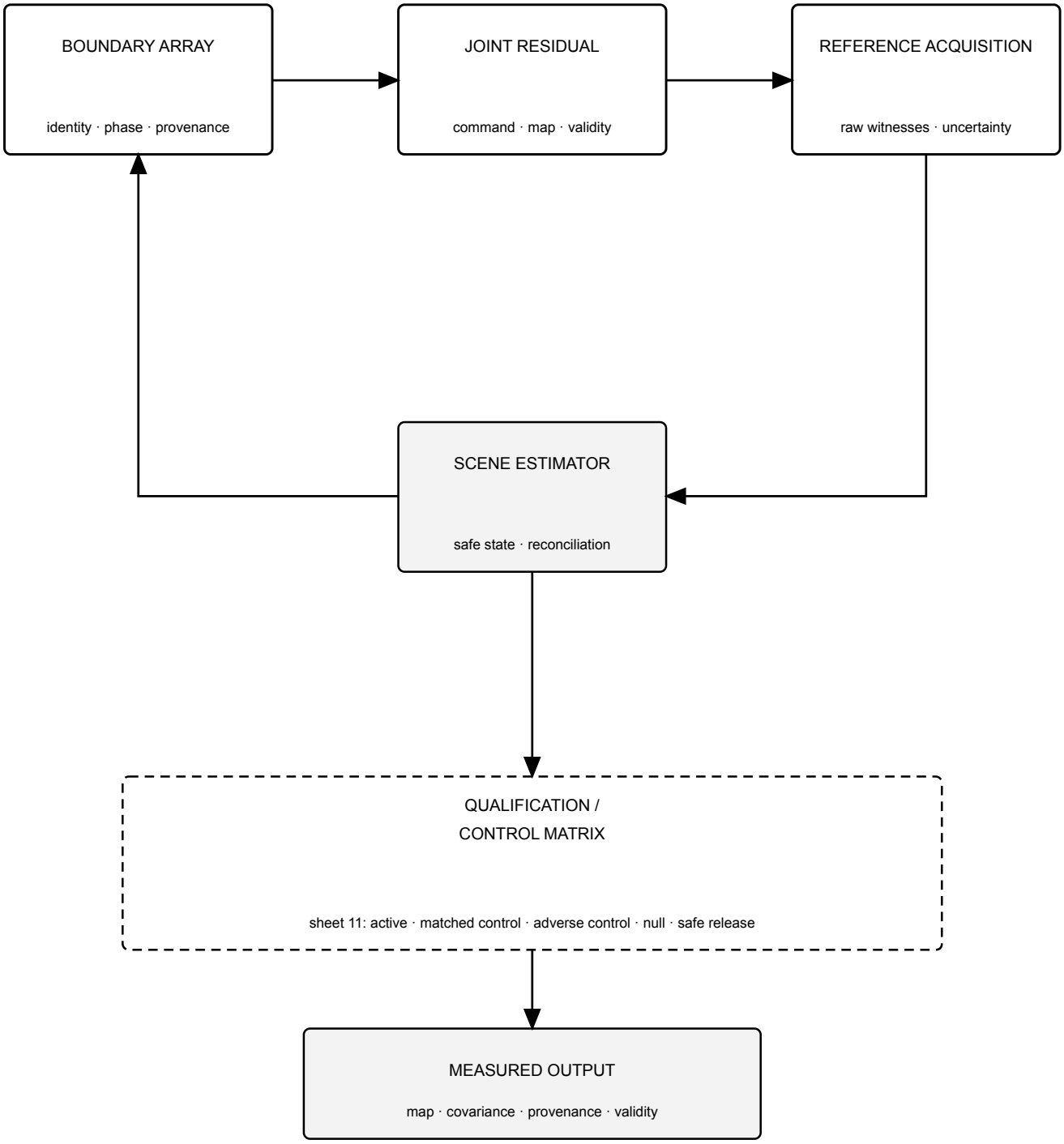


FIG. 11
Graceful-degradation hierarchy and safe-state controller.

Prophetic address-denial selector branch and independent discrimination gates

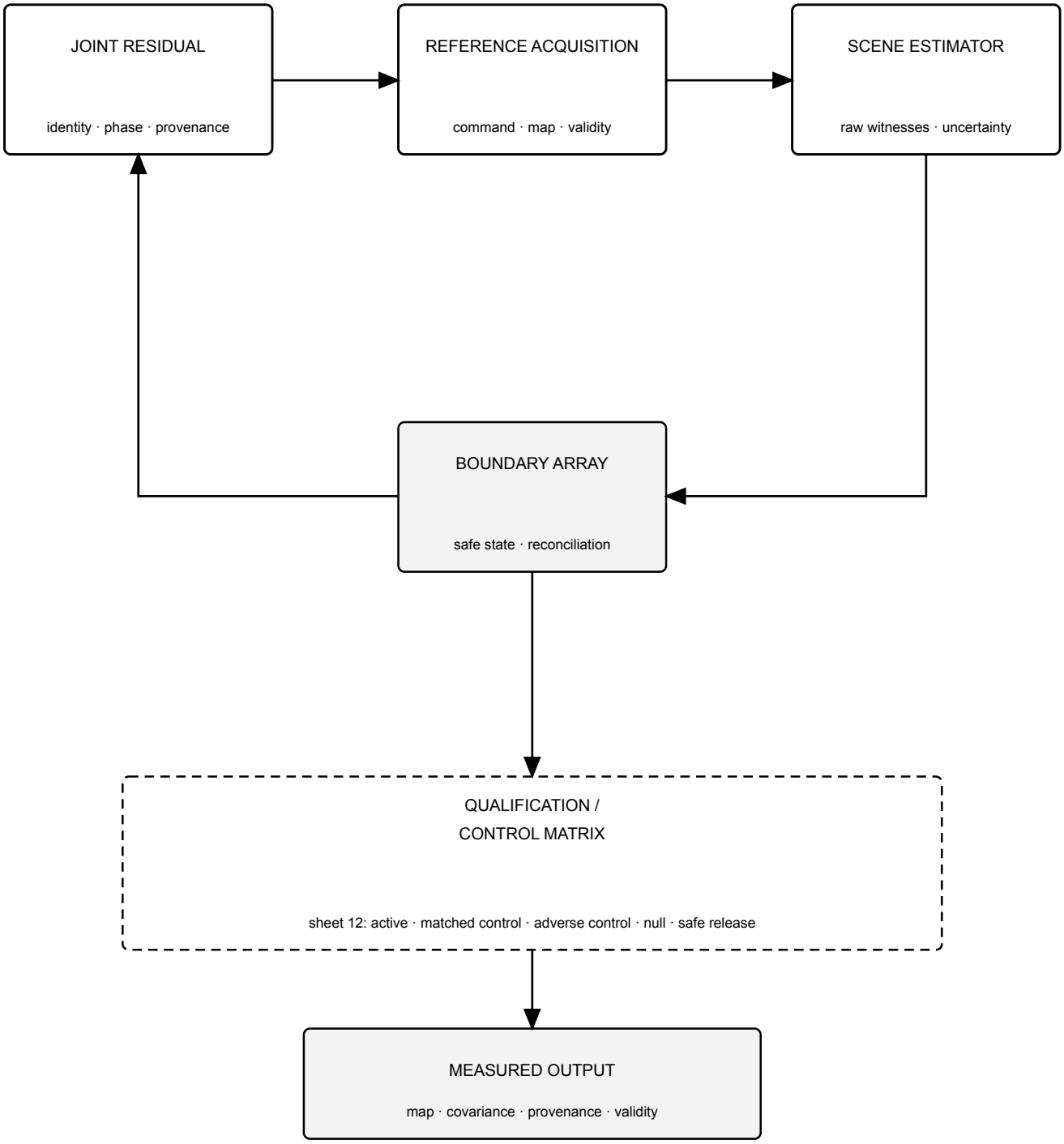


FIG. 12
Prophetic address-denial selector branch and independent discrimination gates.